

DAVID LIANG

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EDUCATION

University of Southern California

Expected Graduation: May 2028

Master of Science in Game Design and Development

- First-Year Graduate Student

Washington University in St. Louis

Graduated: May 2026

Major in Computer Science | Minor in Human-Computer Interaction, and Design

- GPA: 3.84/4.00 | Dean's List all semesters

EXPERIENCE

Washington University Imaging Science

Sept 2025 - May 2026

Research Assistant

- Experimented with creating different games in Unity for the eye tracking process of the OCT scan.
- Collaborated with PhD students in different fields to refine the visual and functional design through weekly meetings and 3D modeling software.

Spectrum (Charter Communications)

May 2025 - Jan 2026

Intern | Part-time Billing Automation Engineer

- Developed automation scripts in C#, Python and PowerShell to streamline internal workflows.
- Built a SharePoint to network share migration tool, reclaiming several terabytes of storage space and reducing infrastructure costs.
- Collaborated with senior engineers to improve maintainability and efficiency of legacy automation systems.
- Currently working on validating and migrating legacy SQL scripts to a different cloud based data platform (Oracle to Snowflake); Collaborating with other engineering teams to update, deploy, and use automated data migration tools.

PERSONAL PROJECTS

Pending Departures – Unity (PC)

July 2026

Solo Project

- Designed an airport operations game where players cross-reference flight records and make departure decisions under pressure.
- Built a progressively complex learning curve, introducing maintenance, weather, and crew requirements through levels, keeping players in the flow channel.
- Observed live player gameplay to identify unclear texts and gaps in decision feedback.

HandSpell - Unity 3D (PC)

May 2025 - Dec 2025

Team Project

- Built a real-time hand-gesture recognition system using Python, MediaPipe, and OSC with Unity, enabling hand gesture spell-casting inputs in a Unity game environment.
- Designed a turn-based, combat system driven by spell types, hand gestures, and puzzle mini games.
- Led the end-to-end prototyping, from paper testing to balancing gesture complexity and pacing.

The Perfect LOOP - Unity 3D (PC)

Aug 2025

Solo Project

- Developed a minimalistic puzzle game about building a single continuous loop from environment clues.
- Received player feedback and refined puzzles, level progression and bugs; played during GMTK Game Jam livestream.

The Curse of Brookings Hall – Unity 3D (PC)

Jan 2025 - May 2026

Team Project

- Led a four-person team from core loop design to final build, coordinating tasks and regular milestones.
- Designed exploration and puzzle-solving layouts using lighting, props and spatial cues to guide players.
- Explored enemy patrol and chase behavior with Unity NavMesh and C#, plus post-processing and lighting.

SKILLS

Technical: Unity; Unreal Engine; C#; Python; C++; HTML; CSS; SQL; Swift; Git; Assembly; Linux; AWS; Blender; Maya; Fusion360; PowerShell; Illustrator; Premiere Pro; Jira; DBever; Microsoft Office

Extra Interests: Video game Design; Interior Design; Sculpturing; Cooking/Baking; Gardening